

SYLVOPAST ROLE-GAME

USERS GUIDE

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Before starting

Two players are required to play to SYLVOPAST role-game, and a PC computer with Excel97 from Microsoft.

Copy the autojeuspeng.xls on your hard disk.

Read very carefully the following game rules

Print a copy of the game and monitoring boards provided at the end of this guide :

- Slide 1 as the map of the forest
- Slides 2 and 3 for the shepherd-player
- Slides 4 to 7 for the forester-player

Open the autojeuspeng.xls file :

- If not available tick the spgame toolbar in the Screen Menu
- run a demo by clicking upon the key button to understand how to play
- run the game by clicking upon the smile button and by answering to the questions

At the end of the game save the file as another file name or reset the game by clicking upon the garbage button

Game rules

Common information

Each plant type corresponds to a basic colour : blue for trees, red for shrubs and yellow for herbs. The forest vegetation is mapped as the combination of plant types expressed by the corresponding combined colour.

A strong dominant wind blows frequently from west to east.

The forest suffers from climatic hazard with dry, mild or rainy conditions.

Three techniques are available to modify vegetation structure :

- S =sowing (adding herbs) costs 100
- D = clearing (getting rid of the shrubs) costs 250
- R = reforesting (planting trees) costs 1000

Shepherd information

The pastoral value of the vegetation depends on vegetation structure and climate (slides 3 and 4). Vegetation dynamics depends on grazing frequency : 3 rounds (consecutive or not) without grazing lead to shrub encroachment (slide 3).

A grazing duty corresponding to the flock size (1 per animal, starting at 250), has to be paid every round.

During each round the shepherd moves his flock on k squares, k corresponding to the flock size divided by 25 (starting at 10) and sums up the square pastoral value to his capital.

He selects the squares according to climate and vegetation (slide 3) and travelling distance.

Any travelling (diagonal forbidden) of more than 3 squares costs 5, and of more than 8 squares costs 10.

At the end of each round, 10 minutes to negotiate an improvement of the forest and to decide to support or not the operations.

At the beginning of a new round, can buy animals ; each lot of 25 animals costs 25 and gives the right to graze a new square.

The shepherd leaves the forest if his capital is lower than 80% of the flock size.

Forester information

Can choose among three goals : to reduce fire hazard, to increase the area of dense forest stands (blue squares) and to increase vegetation diversity.

Fire hazard depends on vegetation structure and grazing (slides 4 and 5).

Fire propagation depends on vegetation structure and grazing (slide 4).

Vegetation dynamics depends on natural succession and can be modified by technical operations such as sowing, clearing, reforestation or grazing (slide 6).

At the beginning of each round collects the grazing duty.

At the end of each round, 10 minutes to negotiate an improvement of the forest according to his primary objectives.

If the global value of the forest has been improved (sum of the 3 criteria must be positive) the forester gets 250.

Stage 1

Meanwhile the forester is planning the best way of improving the forest, the shepherd pays the grazing duty and herds his flock.

10 times, according to the type of climate, the shepherd moves his flock to an ungrazed square, collects the corresponding forage and spends the corresponding travelling.

The forester and the shepherd have 10 minutes to negotiate an improvement of the forest, to agree on some operations and to pay for them.

If after the improvement the global value of the forest increased, the forester gets 250.

The shepherd can increase the flock size.

After five rounds, a fire is randomly set and crosses the forest from West to East according to fire propagation rules.

First balance on the management of the forest according to forestry, livestock and landscape criteria.

You can stop the game at this stage or follow on the second stage.

Stage 2

The forester gets 500 points to recover the forest after the fire.

Meanwhile the forester is planning the best way of improving the forest after the fire, the shepherd pays the grazing duty and herds his flock.

10 times (or more if the flock size increased), according to the type of climate, the shepherd moves his flock to an ungrazed square, collects the corresponding forage and spends the corresponding travelling.

The forester and the shepherd have 10 minutes to negotiate an improvement of the forest, to agree on some operations and to pay for them

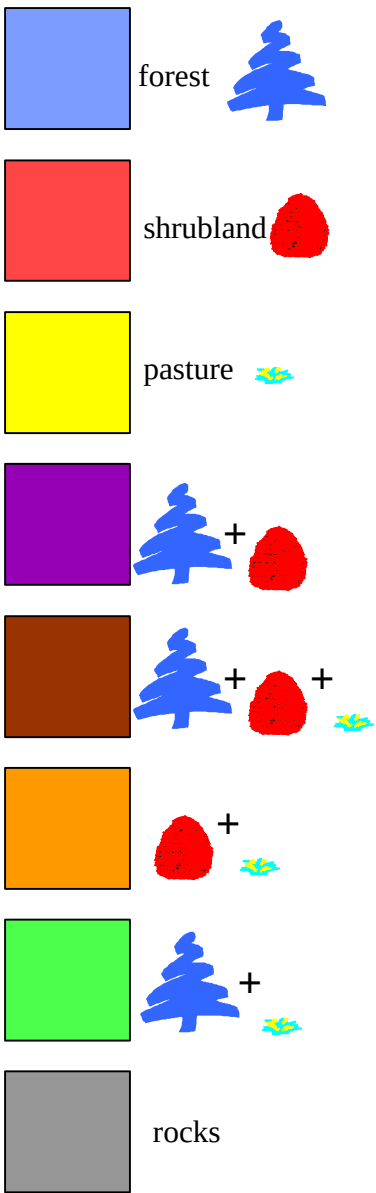
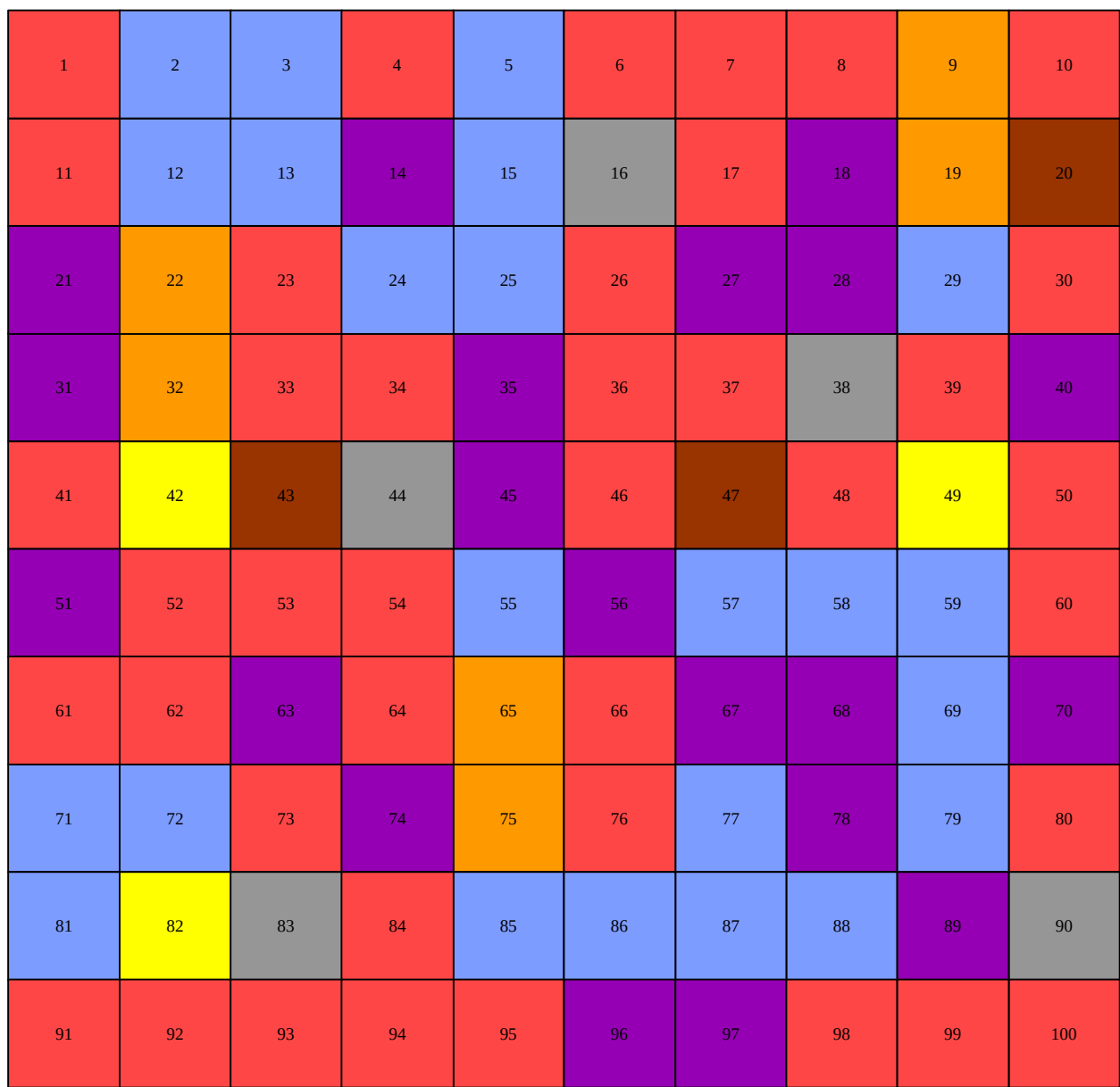
If after the improvement the global value of the forest increased, the forester gets 250.

The shepherd can increase the flock size.

After five rounds, a fire is randomly set in a shrubby square and crosses the forest from West to East according to fire propagation rules.

Final balance on the management of the forest according to forestry, livestock and landscape criteria

Please send a copy of your game to my e-mail adress.



pastoral value according to climate

D	M	R
15	30	45
10	15	15
5	15	20
30	50	40

dynamics and grazing

G	NG
	3
	3
	3
	3

	forest 
	shrubla 
	pasture 
	 + 
	 +  + 
	 + 
	 + 
	rocks

shepherd monitoring board

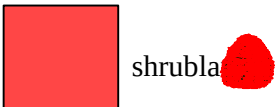
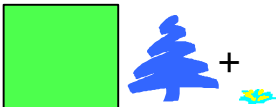
1 low
 2 medium
 3 high
 grazing forbidden in this plot

PV			
FV			

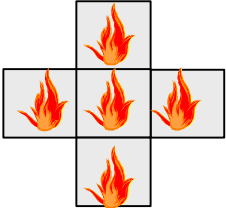
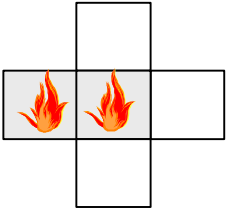
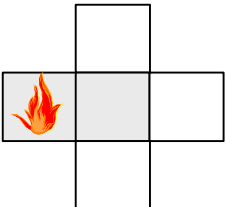
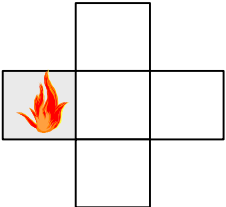
value	herd	capital
15	250	500

$$\text{value} = \sum (\text{VP} * \text{FV})$$

fire hazard according to grazing

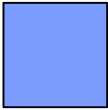
G		NG
3		3
2		3
2		3
		2
2		2
0		1
0		1
		0

and fire propagation

high	3	
medium	2	
low	1	
null	0	

fire hazard in the forest

2	2	2	2	2	2	2	2	3	2
2	2	2	3	2	0	2	3	3	3
3	3	2	2	2	2	3	3	2	2
3	3	2	2	3	2	2	0	2	3
2	1	3	0	3	2	3	2	1	2
3	2	2	2	2	3	2	2	2	2
2	2	3	2	3	2	3	3	2	3
2	2	2	3	3	2	2	3	2	2
2	1	0	2	2	2	2	2	3	0
2	2	2	2	2	3	3	2	2	2



dense forest



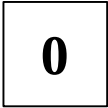
high



medium

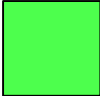
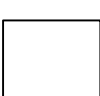


low



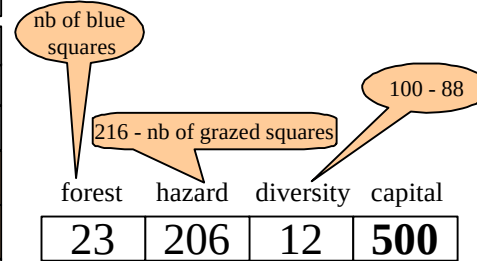
null

vegetation dynamics according to operations and grazing

		operations				grazing	
		D	S	D+S	R	G	NG
	forest 						
	shrubl 						
	pasture 						
	 + 						
	 +  + 						
	 + 						
	 + 						
	rocks						

monitoring board of the forester

	0	2	2	1	3	3	3	1	Σ
FR	0	2	2	1	3	3	3	1	
obj	12	13	13	12	13	12	12	13	
FV	5	23	40	3	20	3	6	0	
FV-obj	7	10	27	9	7	9	6	13	88
FR*FV	0	46	80	3	60	9	18	0	216
FV									
FV-obj									
FR*FV									
FV									
FV-obj									
FR*FV									
FV									
FR-obj									
RI*FV									
FV									
FV-obj									
FR*FV									
FV									
FV-obj									
FR*FV									



forest hazard diversity capital

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